The Sine–Kumaraswamy Distribution and Its GAMLSS Model for Unit-Interval Data

Abstract: *This paper introduces the Sine–Kumaraswamy (SKW) distribution and its corresponding regression formulation, the GAMLSS SKW (GSKW), for modeling continuous data on the unit interval* $(0, 1)$ *without probability mass at the boundaries. The proposed framework provides a flexible probabilistic specification for rates and proportions. The construction is based on the Sine-G generator applied to the Kumaraswamy distribution, yielding a new two-parameter distribution with shape parameters* δ *and* σ*, capable of accommodating a wide range of density shapes. The associated GSKW model is developed within the Generalized Additive Models for Location, Scale and Shape (GAMLSS) framework, allowing both parameters to be modeled as functions of covariates. Inference relies on maximum likelihood estimation (MLE), and the finite-sample behavior of the estimators is examined through Monte Carlo simulations, supporting their consistency and asymptotic normality. The empirical performance of the SKW distribution is illustrated using family planning data, where it provides improved fit relative to competing models, including the Beta, Kumaraswamy, Unit Weibull, Unit Burr XII, Unit Gompertz, and Unit Topp–Leone distributions. In addition, the GSKW model is applied to age dependency ratio data from* 142 *countries.*

Keywords: *Unit Distribution; Sine-Generated Family; GAMLSS Model; Maximum Likelihood Estimation; Monte Carlo Simulation.*

Monte carlo Simulations

A comprehensive Monte Carlo simulation study was performed to investigate the finite-sample properties of parameter estimators for the SKW distribution and the proposed GSKW model. The simulation was based on $10{,}000$ replications across a range of sample sizes $n \in \{50, 100, 200, 400\}$. The performance of the maximum likelihood estimators (MLEs) was systematically evaluated based on their mean, bias, standard deviation (SD), and mean squared error (MSE), as well as their skewness (SC) and kurtosis (KC) coefficients. Additionally, the empirical coverage probability (CP) was computed to assess the performance of the 95% confidence intervals.was used.

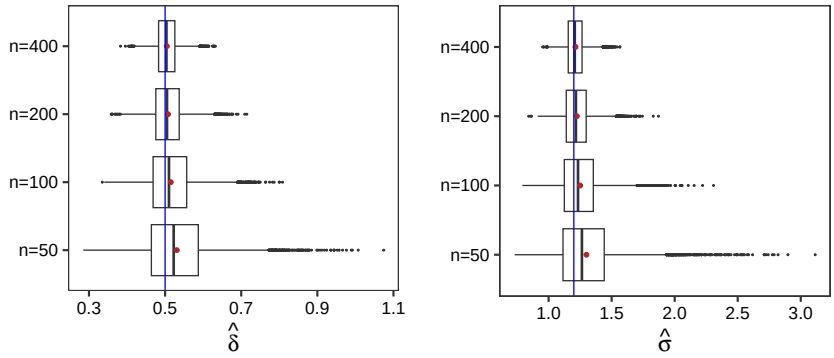
28 ***SKW distribution - Scenario:*** $\delta = 0.50$ ***and*** $\sigma = \{1.20, 2.20, 3.20\}$

Table 1: Monte Carlo simulation results for the MLE of the parameters of the SKW(δ, σ) distribution, considering $\delta = 0.50$ and $\sigma \in \{1.20, 2.20, 3.20\}$, and sample sizes $n \in \{50, 100, 200, 400\}$.

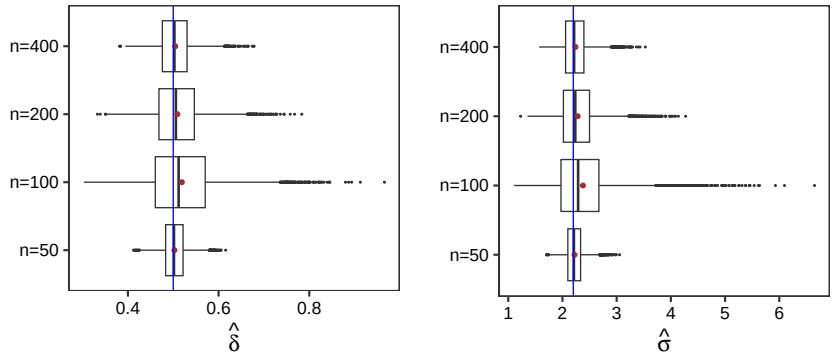
Parameter	n	Mean	Bias	SD	MSE	SC	KS	CP
Scenario: $\delta = 0.50$ and $\sigma = 1.20$								
δ	50	0.5235	0.0235	0.0955	0.0097	0.6181	3.7552	0.9521
	100	0.5112	0.0112	0.0652	0.0044	0.4060	3.2873	0.9512
	200	0.5060	0.0060	0.0454	0.0021	0.3307	3.1198	0.9531
	400	0.5030	0.0030	0.0322	0.0010	0.2805	3.2807	0.9488
σ	50	1.2767	0.0767	0.2630	0.0751	1.0319	5.1206	0.9581
	100	1.2365	0.0365	0.1700	0.0302	0.6309	3.7230	0.9580
	200	1.2186	0.0186	0.1159	0.0138	0.5026	3.5884	0.9594
	400	1.2087	0.0087	0.0812	0.0067	0.3111	3.1635	0.9503
Scenario: $\delta = 0.50$ and $\sigma = 2.20$								
δ	50	0.5196	0.0196	0.0828	0.0072	0.5094	3.4743	0.9525
	100	0.5105	0.0105	0.0585	0.0035	0.4280	3.3020	0.9475
	200	0.5049	0.0049	0.0401	0.0016	0.2563	3.0162	0.9496
	400	0.5018	0.0018	0.0277	0.0008	0.1628	3.0763	0.9544
σ	50	2.3856	0.1856	0.5851	0.3768	1.2230	5.8098	0.9631
	100	2.2902	0.0902	0.3833	0.1550	0.8498	4.3964	0.9536
	200	2.2425	0.0425	0.2564	0.0675	0.5460	3.4892	0.9544
	400	2.2172	0.0172	0.1736	0.0304	0.3656	3.2074	0.9554
Scenario: $\delta = 0.50$ and $\sigma = 3.20$								
δ	50	0.5171	0.0171	0.0778	0.0063	0.5300	3.4352	0.9526
	100	0.5083	0.0083	0.0531	0.0029	0.4019	3.3367	0.9566
	200	0.5053	0.0053	0.0376	0.0014	0.2046	3.0111	0.9517
	400	0.5024	0.0024	0.0267	0.0007	0.1863	3.0251	0.9513
σ	50	3.5010	0.3010	0.9657	1.0233	1.4216	6.9619	0.9631
	100	3.3454	0.1454	0.6027	0.3844	0.9636	5.2459	0.9641
	200	3.2798	0.0798	0.4108	0.1752	0.5512	3.4537	0.9554
	400	3.2379	0.0379	0.2855	0.0830	0.4332	3.3848	0.9539

Source: From the authors (2026).

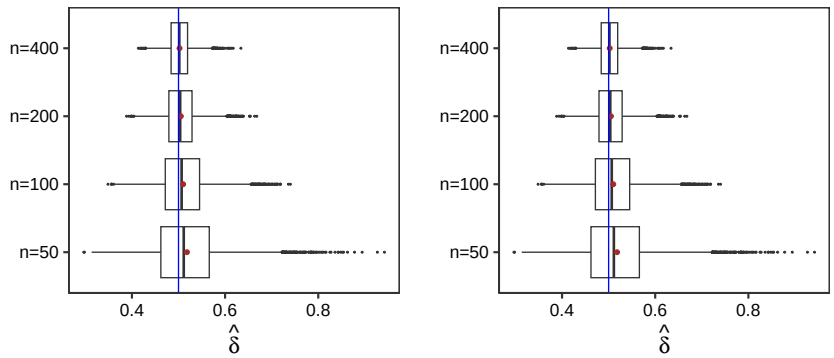
Figure 1: Boxplots of the Monte Carlo simulation results for the MLE of the parameters of the SKW(δ, σ) distribution, considering $\delta = 0.50$ and $\sigma \in \{1.20, 2.20, 3.20\}$, and sample sizes $n \in \{50, 100, 200, 400\}$.



(a) Scenario: $\delta = 0.5$ and $\sigma = 1.2$



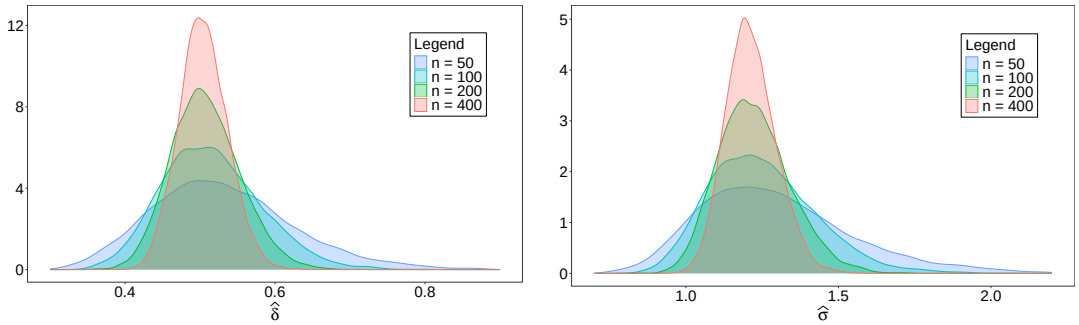
(b) Scenario: $\delta = 0.5$ and $\sigma = 2.2$



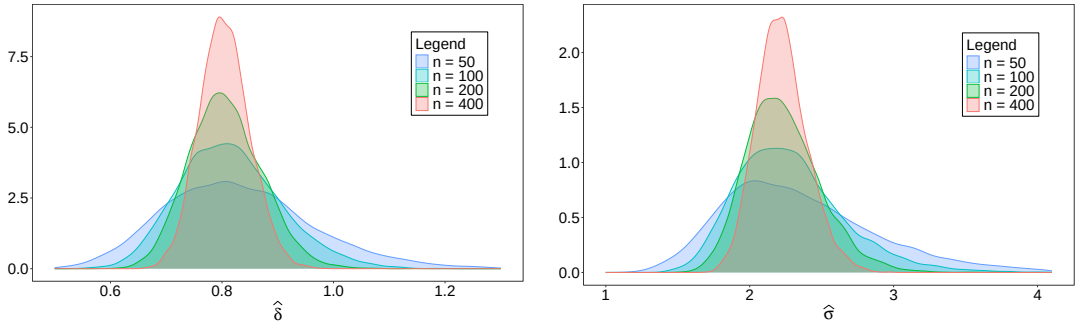
(c) Scenario: $\delta = 0.5$ and $\sigma = 3.2$

Source: From the authors (2026).
Legend: -: true parameter value; •: mean of the 10,000 replications.

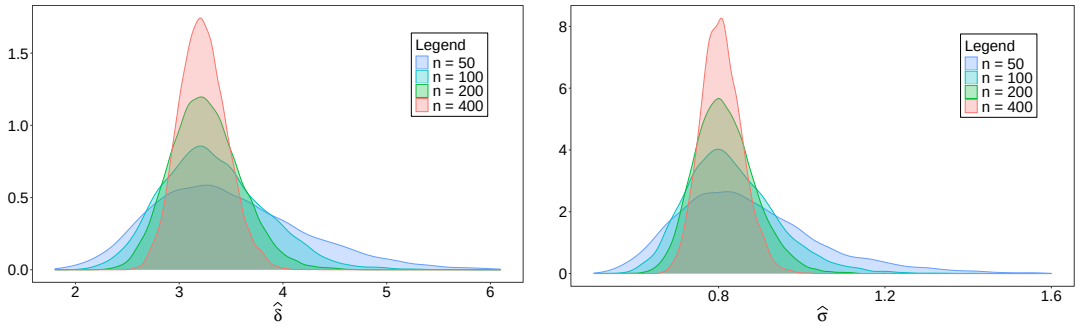
Figure 2: Sampling densities of the Monte Carlo simulation results for the MLE of the parameters of the SKW(δ, σ) distribution, considering $\delta = 0.50$ and $\sigma \in \{1.20, 2.20, 3.20\}$, and sample sizes $n \in \{50, 100, 200, 400\}$.



(a) Scenario: $\delta = 0.5$ and $\sigma = 1.2$



(b) Scenario: $\delta = 0.8$ and $\sigma = 2.2$



(c) Scenario: $\delta = 3.2$ and $\sigma = 0.8$

Source: From the authors (2026).

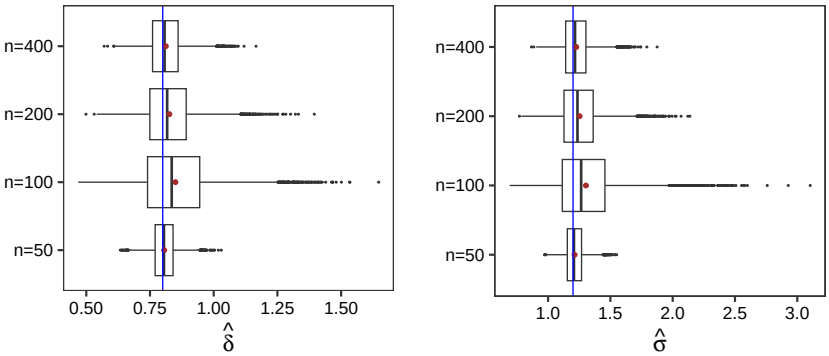
29 ***SKW distribution - Scenario:*** $\delta = 0.80$ ***and*** $\sigma = \{1.20, 2.20, 3.20\}$

Table 2: Monte Carlo simulation results for the MLE of the parameters of the SKW(δ, σ) distribution, considering $\delta = 0.80$ and $\sigma \in \{1.20, 2.20, 3.20\}$, and sample sizes $n \in \{50, 100, 200, 400\}$.

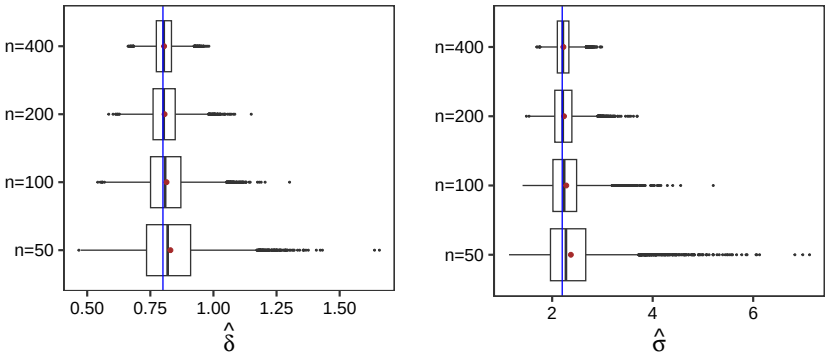
Parameter	n	Mean	Bias	SD	MSE	SC	KC	CP
Scenario: $\delta = 0.80$ and $\sigma = 1.20$								
δ	50	0.8499	0.0499	0.1539	0.0262	0.5467	3.4248	0.9510
	100	0.8262	0.0262	0.1075	0.0122	0.4496	3.3119	0.9461
	200	0.8122	0.0122	0.0747	0.0057	0.3337	3.0903	0.9458
	400	0.8060	0.0060	0.0519	0.0027	0.1756	3.0136	0.9471
σ	50	1.3038	0.1038	0.2674	0.0823	0.9296	4.5747	0.9613
	100	1.2532	0.0532	0.1738	0.0330	0.5710	3.4999	0.9594
	200	1.2265	0.0265	0.1209	0.0153	0.4525	3.2767	0.9505
	400	1.2129	0.0129	0.0834	0.0071	0.2755	2.9601	0.9504
Scenario: $\delta = 0.80$ and $\sigma = 2.20$								
δ	50	0.8297	0.0297	0.1331	0.0186	0.5389	3.6058	0.9486
	100	0.8143	0.0143	0.0902	0.0083	0.3925	3.2193	0.9538
	200	0.8065	0.0065	0.0651	0.0043	0.2864	3.1580	0.9480
	400	0.8046	0.0046	0.0450	0.0020	0.1780	3.0345	0.9505
σ	50	2.3718	0.1718	0.5833	0.3698	1.3571	6.9318	0.9605
	100	2.2812	0.0812	0.3738	0.1463	0.8841	4.6675	0.9580
	200	2.2357	0.0357	0.2590	0.0684	0.6134	3.8080	0.9467
	400	2.2238	0.0238	0.1760	0.0315	0.3670	3.2076	0.9495
Scenario: $\delta = 0.80$ and $\sigma = 3.20$								
δ	50	0.8295	0.0295	0.1251	0.0165	0.5342	3.5923	0.9520
	100	0.8159	0.0159	0.0869	0.0078	0.3675	3.2497	0.9482
	200	0.8072	0.0072	0.0605	0.0037	0.3032	3.1407	0.9495
	400	0.8038	0.0038	0.0422	0.0018	0.1718	3.0761	0.9506
σ	50	3.5126	0.3126	0.9690	1.0366	1.4045	6.8603	0.9633
	100	3.3567	0.1567	0.6139	0.4014	0.8854	4.4484	0.9575
	200	3.2758	0.0758	0.4148	0.1778	0.6324	3.6841	0.9532
	400	3.2391	0.0391	0.2854	0.0830	0.4080	3.2961	0.9512

Source: From the authors (2026).

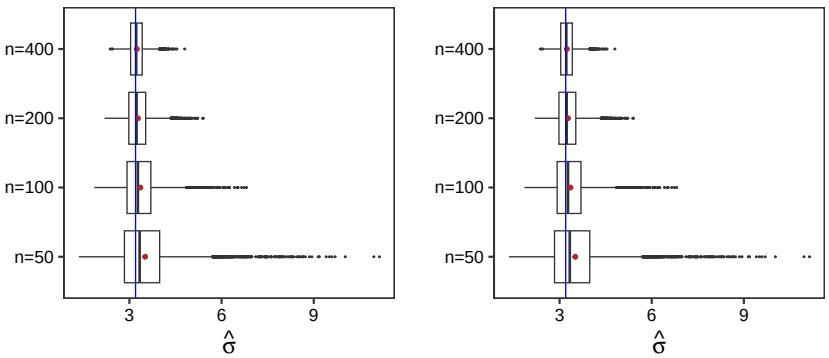
Figure 3: Boxplots of the Monte Carlo simulation results for the MLE of the parameters of the SKW(δ, σ) distribution, considering $\delta = 0.80$ and $\sigma \in \{1.20, 2.20, 3.20\}$, and sample sizes $n \in \{50, 100, 200, 400\}$.



(a) Scenario: $\delta = 0.8$ and $\sigma = 1.2$



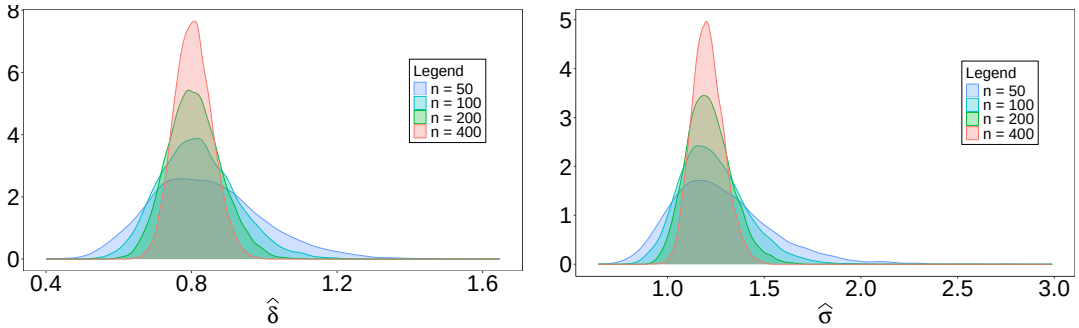
(b) Scenario: $\delta = 0.8$ and $\sigma = 2.2$



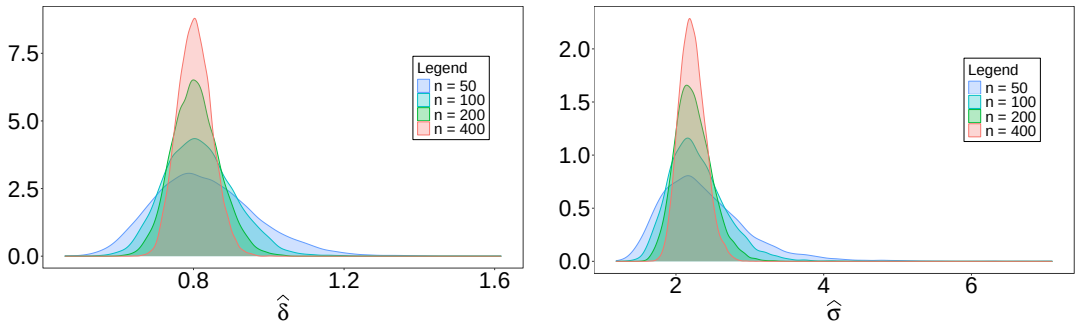
(c) Scenario: $\delta = 0.8$ and $\sigma = 3.2$

Source: From the authors (2026).
Legend: -: true parameter value; •: mean of the 10,000 replications.

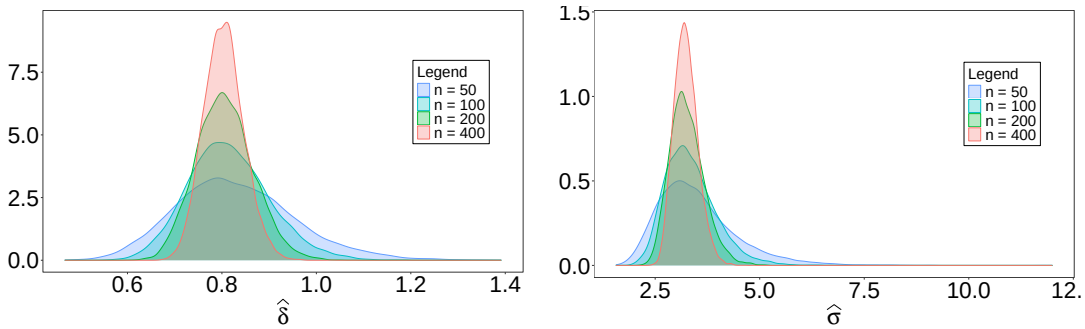
Figure 4: Sampling densities of the Monte Carlo simulation results for the MLE of the parameters of the SKW(δ, σ) distribution, considering $\delta = 0.80$ and $\sigma \in \{1.20, 2.20, 3.20\}$, and sample sizes $n \in \{50, 100, 200, 400\}$.



(a) Scenario: $\delta = 0.8$ and $\sigma = 1.2$



(b) Scenario: $\delta = 0.8$ and $\sigma = 2.2$



(c) Scenario: $\delta = 0.8$ and $\sigma = 3.2$

Source: From the authors (2026).

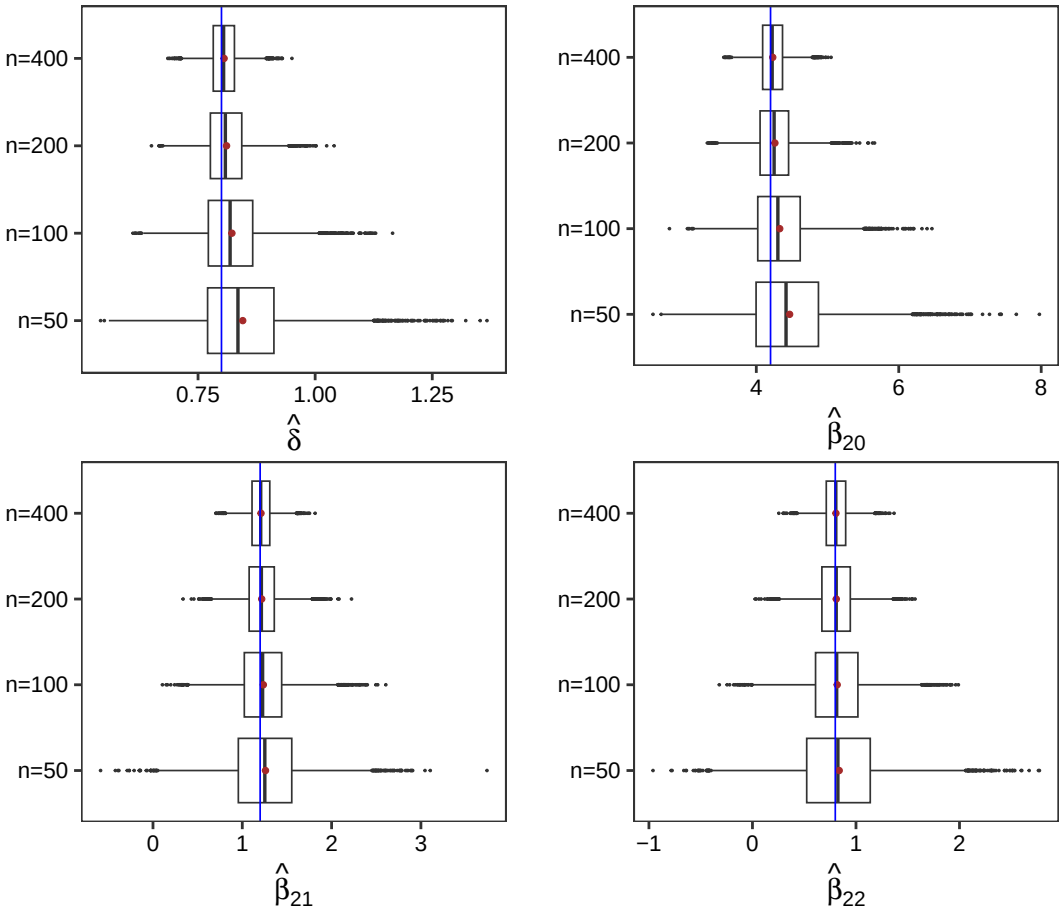
30 ***GSKW model - Scenario:*** $\delta = 0.80$ ***and*** $\boldsymbol{\beta}_2 = (4.2, 1.2, 0.8)$

Table 3: Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\delta = 0.80$ and $\boldsymbol{\beta}_2 = (4.2, 1.2, 0.8)$, and sample sizes $n \in \{50, 100, 200, 400\}$.

Parameter	n	Mean	Bias	SD	MSE	SC	KC	CP
δ	50	0.8454	0.0454	0.1064	0.0134	0.5421	3.5175	0.9399
	100	0.8223	0.0223	0.0715	0.0056	0.4038	3.4093	0.9390
	200	0.8109	0.0109	0.0495	0.0026	0.2354	3.0678	0.9425
	400	0.8056	0.0056	0.0341	0.0012	0.1612	3.0578	0.9468
β_{20}	50	4.4680	0.2680	0.6656	0.5148	0.4997	3.4213	0.9344
	100	4.3291	0.1291	0.4487	0.2180	0.3835	3.3550	0.9406
	200	4.2614	0.0614	0.2984	0.0928	0.2612	3.2044	0.9450
	400	4.2310	0.0310	0.2057	0.0433	0.1680	3.0299	0.9445
β_{21}	50	1.2599	0.0599	0.4482	0.2045	0.1383	3.2442	0.9437
	100	1.2358	0.0358	0.3126	0.0990	0.1635	3.1675	0.9433
	200	1.2174	0.0174	0.2152	0.0466	0.0921	3.1418	0.9457
	400	1.2094	0.0094	0.1485	0.0221	0.0709	2.9360	0.9504
β_{22}	50	0.8379	0.0379	0.4673	0.2198	0.1511	3.1920	0.9419
	100	0.8186	0.0186	0.3058	0.0939	0.1149	3.0946	0.9431
	200	0.8095	0.0095	0.2061	0.0426	0.0368	3.0300	0.9456
	400	0.8063	0.0063	0.1396	0.0195	0.0135	2.9703	0.9505

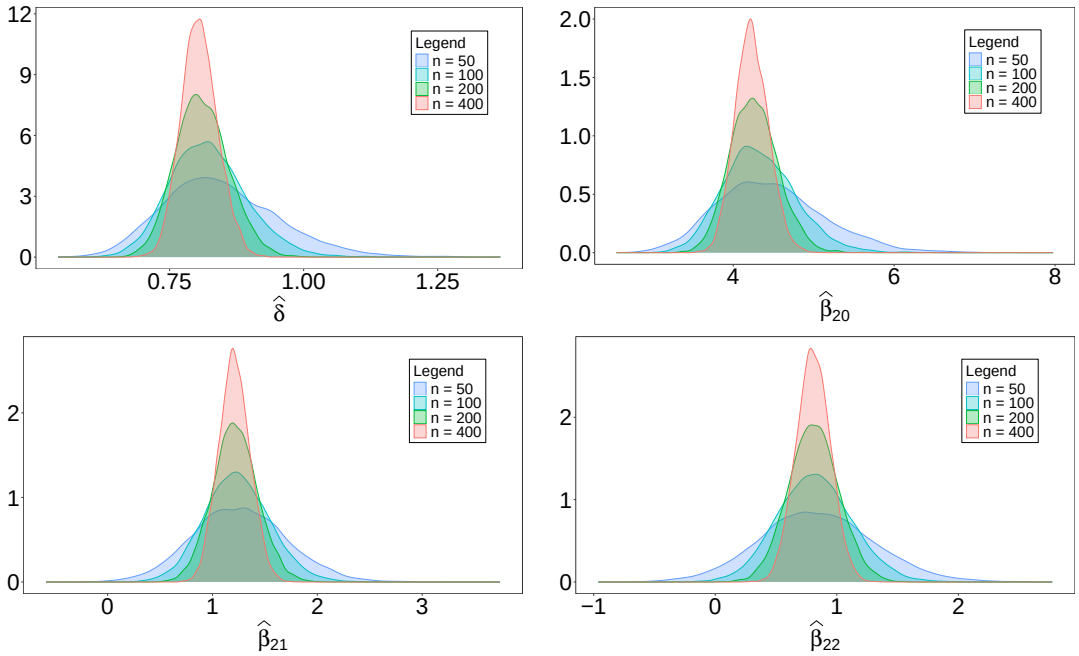
Source: From the authors (2026).

Figure 5: Boxplots of the Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\delta = 0.8$ and $\boldsymbol{\beta}_2 = (4.2, 1.2, 0.8)$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).
Legend: ▬: true parameter value; •: mean of the 10,000 replications.

Figure 6: Sampling densities of the Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\delta = 0.8$ and $\boldsymbol{\beta}_2 = (4.2, 1.2, 0.8)$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).

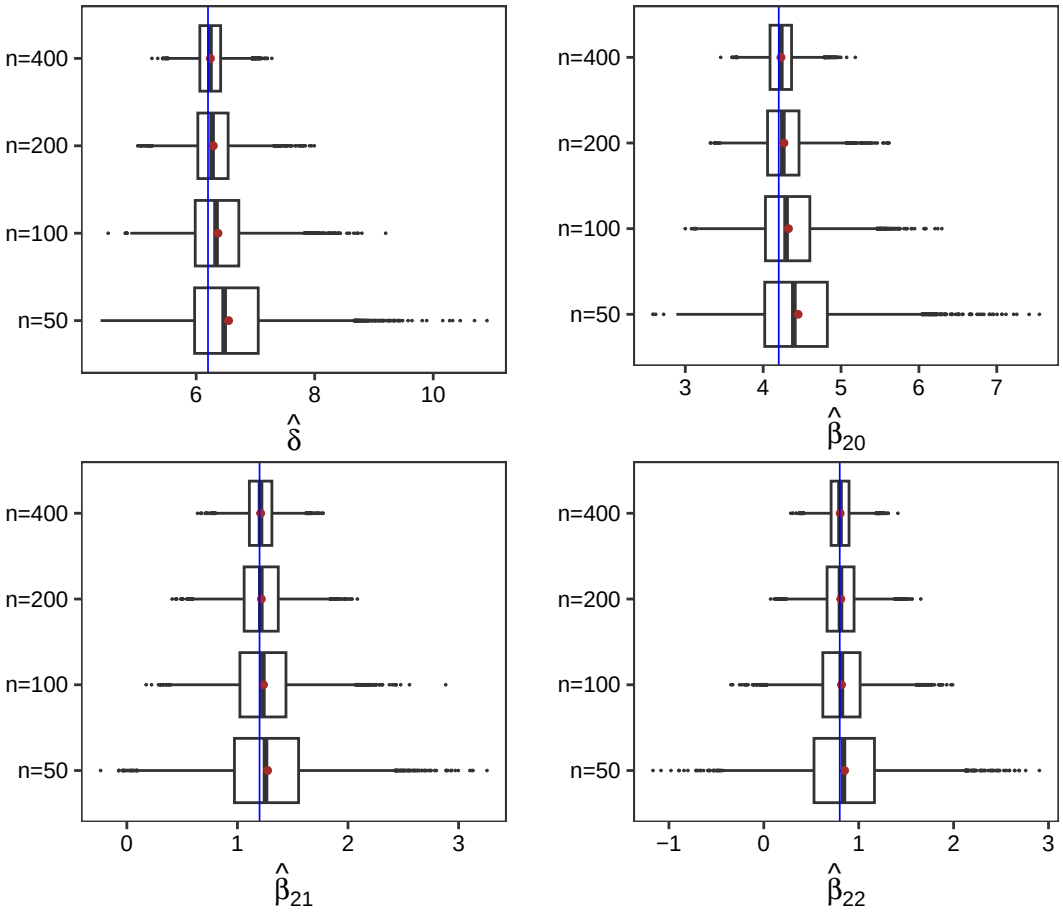
31 ***GSKW model - Scenario:*** $\delta = 6.20$ ***and*** $\boldsymbol{\beta}_2 = (4.2, 1.2, 0.8)$

Table 4: Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\delta = 6.20$ and $\boldsymbol{\beta}_2 = (4.2, 1.2, 0.8)$, and sample sizes $n \in \{50, 100, 200, 400\}$.

Parameter	n	Mean	Bias	SD	MSE	SC	KC	CP
δ	50	6.5479	0.3479	0.8067	0.7719	0.5120	3.4323	0.9443
	100	6.3689	0.1689	0.5512	0.3323	0.3750	3.3457	0.9448
	200	6.2897	0.0897	0.3810	0.1532	0.2767	3.2117	0.9480
	400	6.2415	0.0415	0.2634	0.0711	0.1556	3.0361	0.9485
β_{20}	50	4.4491	0.2491	0.6156	0.4410	0.5299	3.5071	0.9401
	100	4.3250	0.1250	0.4291	0.1998	0.3683	3.2252	0.9441
	200	4.2668	0.0668	0.2979	0.0932	0.2974	3.2040	0.9473
	400	4.2292	0.0292	0.2027	0.0419	0.1610	3.0373	0.9499
β_{21}	50	1.2732	0.0732	0.4361	0.1955	0.2393	3.2499	0.9426
	100	1.2363	0.0363	0.3120	0.0986	0.1996	3.1850	0.9436
	200	1.2166	0.0166	0.2260	0.0513	0.1045	3.0463	0.9452
	400	1.2104	0.0104	0.1513	0.0230	0.0515	3.0462	0.9464
β_{22}	50	0.8513	0.0513	0.4820	0.2349	0.1252	3.2241	0.9326
	100	0.8193	0.0193	0.2978	0.0891	0.0526	3.1652	0.9435
	200	0.8115	0.0115	0.2113	0.0448	0.0678	3.0511	0.9466
	400	0.8047	0.0047	0.1415	0.0200	0.0788	3.0433	0.9448

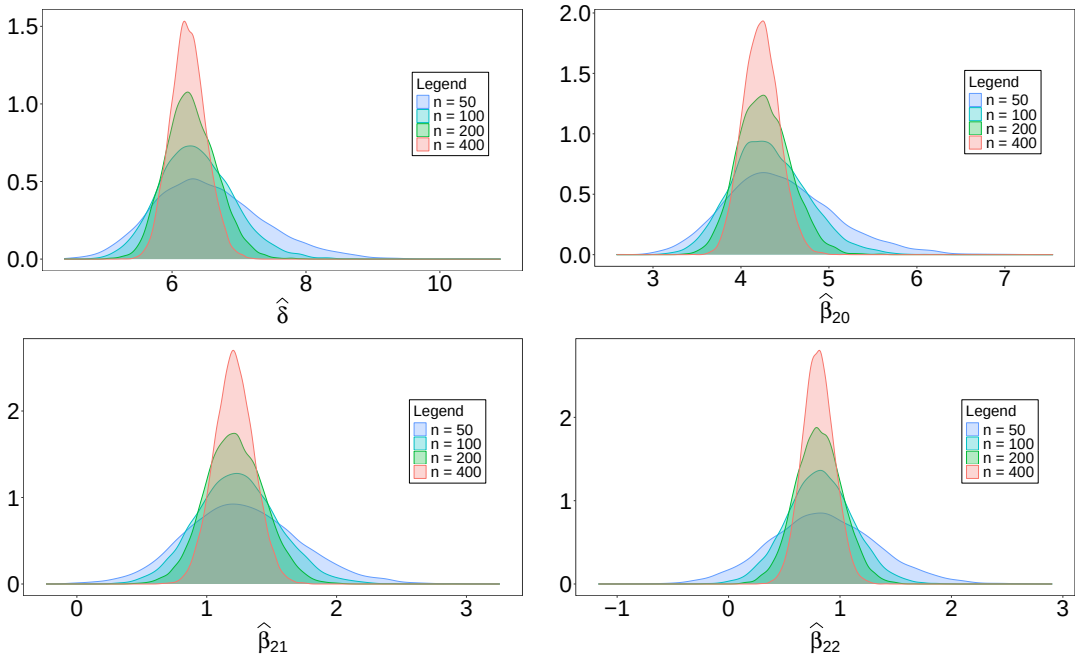
Source: From the authors (2026).

Figure 7: Boxplots of the Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\delta = 6.2$ and $\boldsymbol{\beta}_2 = (4.2, 1.2, 0.8)$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).
Legend: ▬: true parameter value; •: mean of the 10,000 replications.

Figure 8: Sampling densities of the Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\delta = 6.2$ and $\boldsymbol{\beta}_2 = (4.2, 1.2, 0.8)$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).

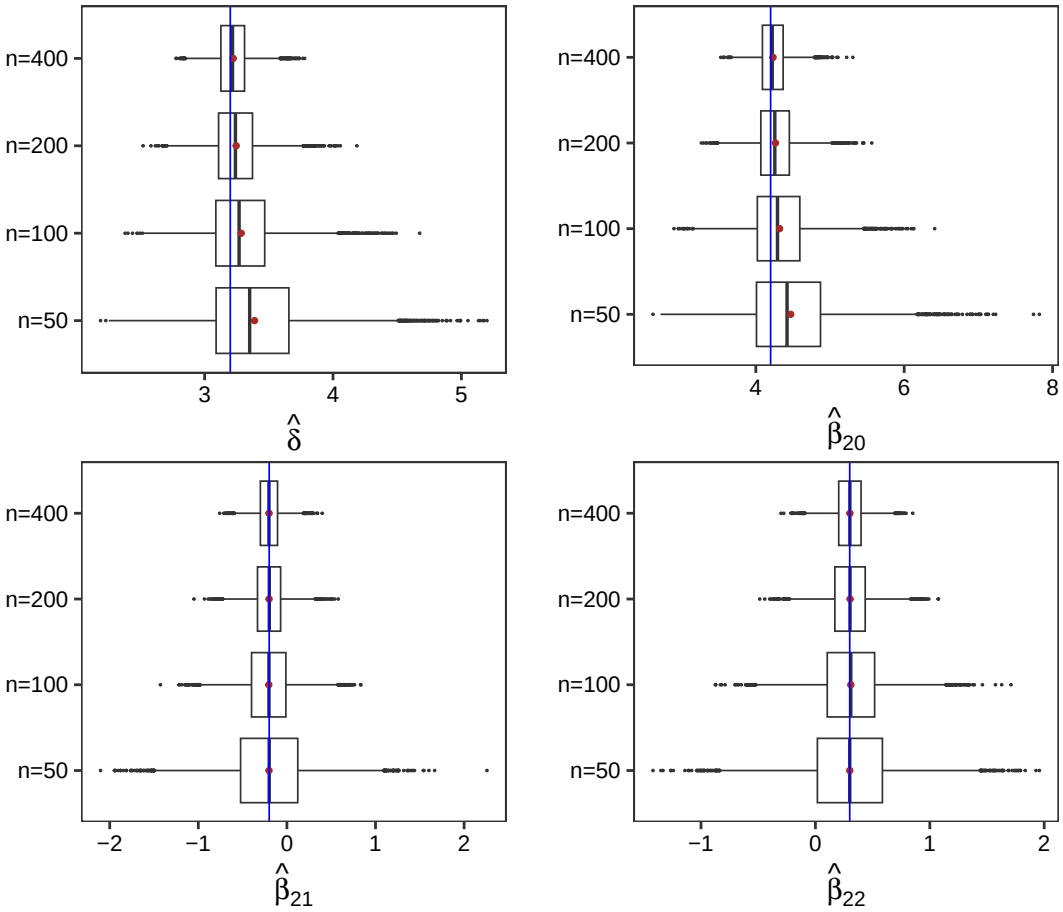
32 ***GSKW model - Scenario:*** $\delta = 3.20$ ***and*** $\boldsymbol{\beta}_2 = (4.2, -0.2, 0.3)$

Table 5: Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\delta = 3.20$ and $\boldsymbol{\beta}_2 = (4.2, -0.2, 0.3)$, and sample sizes $n \in \{50, 100, 200, 400\}$.

Parameter	n	Mean	Bias	SD	MSE	SC	KC	CP
δ	50	3.3890	0.1890	0.4254	0.2167	0.4573	3.2358	0.9434
	100	3.2851	0.0851	0.2867	0.0895	0.3482	3.2760	0.9422
	200	3.2458	0.0458	0.1994	0.0418	0.2377	3.1545	0.9448
	400	3.2225	0.0225	0.1381	0.0196	0.2288	3.1632	0.9468
β_{20}	50	4.4704	0.2704	0.6546	0.5016	0.5001	3.4225	0.9385
	100	4.3204	0.1204	0.4324	0.2014	0.3422	3.2539	0.9410
	200	4.2649	0.0649	0.2906	0.0886	0.2193	3.1769	0.9437
	400	4.2326	0.0326	0.2087	0.0446	0.2538	3.2269	0.9449
β_{21}	50	-0.2020	-0.0020	0.4853	0.2355	-0.0179	3.2239	0.9395
	100	-0.2042	-0.0042	0.2862	0.0819	-0.0069	3.0648	0.9405
	200	-0.2013	-0.0013	0.1993	0.0397	-0.0299	3.0637	0.9444
	400	-0.2018	-0.0018	0.1428	0.0204	0.0512	3.0469	0.9461
β_{22}	50	0.3002	0.0002	0.4274	0.1826	-0.0061	3.2224	0.9359
	100	0.3098	0.0098	0.3129	0.0980	0.0337	3.1651	0.9395
	200	0.3035	0.0035	0.2012	0.0405	0.0289	3.0916	0.9457
	400	0.3018	0.0018	0.1444	0.0209	-0.0038	3.0516	0.9479

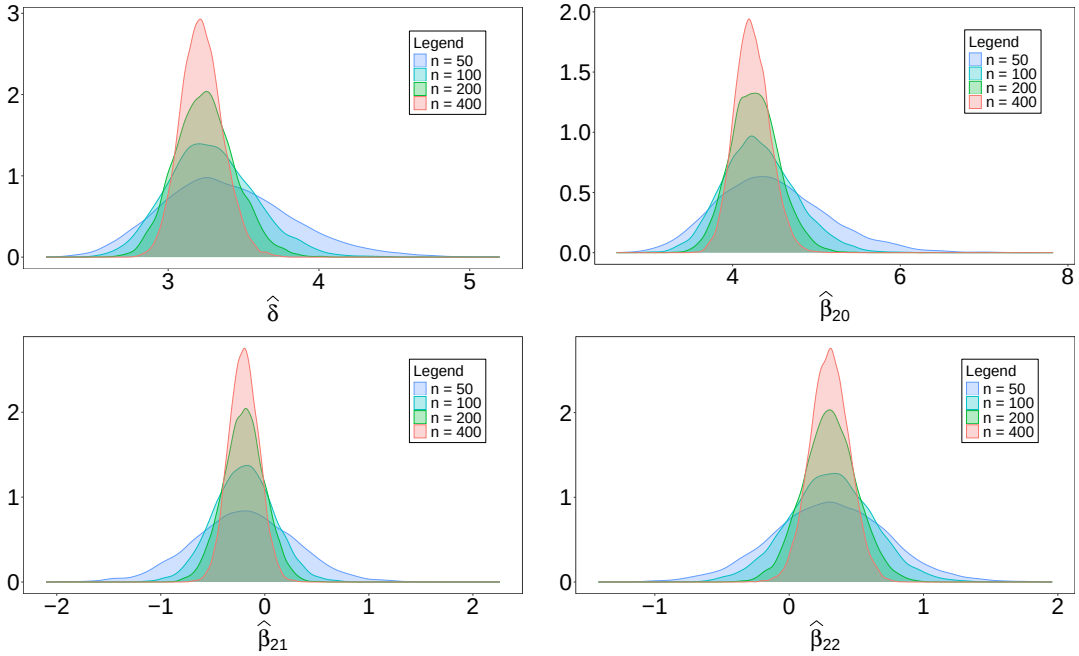
Source: From the authors (2026).

Figure 9: Boxplots of the Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\delta = 3.2$ and $\boldsymbol{\beta}_2 = (4.2, -0.2, 0.3)$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).
Legend: ▬: true parameter value; •: mean of the 10,000 replications.

Figure 10: Sampling densities of the Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\delta = 3.2$ and $\boldsymbol{\beta}_2 = (4.2, -0.2, 0.3)$, and sample sizes $n \in \{50, 100, 200, 400\}$



Source: From the authors (2026).

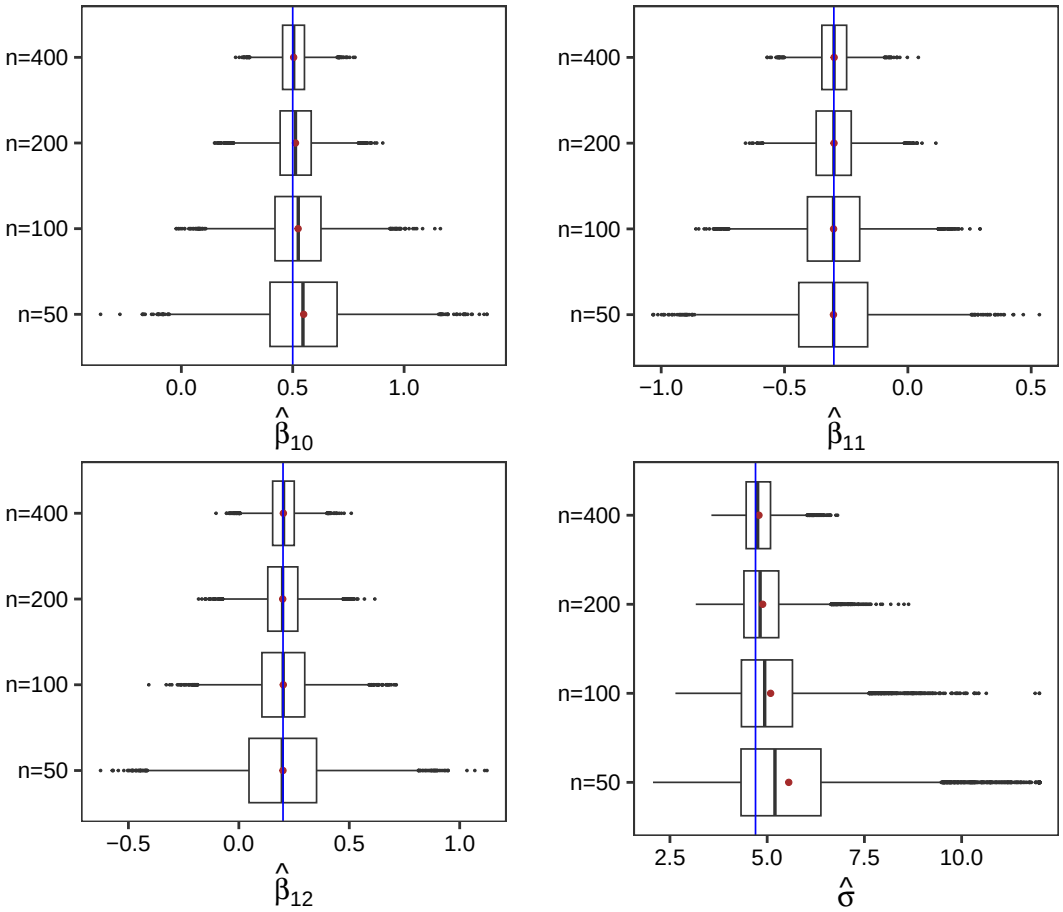
33 ***GSKW model - Scenario:*** $\boldsymbol{\beta}_1 = (0.5, -0.3, 0.2)$ ***and*** $\sigma = 4.7$

Table 6: Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (0.5, -0.3, 0.2)$ and $\sigma = 4.7$, and sample sizes $n \in \{50, 100, 200, 400\}$.

Parameter	n	Mean	Bias	SD	MSE	SC	KC	CP
β_{10}	50	0.5491	0.0491	0.2241	0.0526	0.0423	2.9741	0.9276
	100	0.5241	0.0241	0.1535	0.0241	0.0028	3.0468	0.9411
	200	0.5125	0.0125	0.1038	0.0109	-0.0263	2.9816	0.9486
	400	0.5041	0.0041	0.0719	0.0052	0.0170	2.9337	0.9527
β_{11}	50	-0.3016	-0.0016	0.2075	0.0431	-0.0012	3.0258	0.9385
	100	-0.3011	-0.0011	0.1593	0.0254	0.0099	3.0097	0.9435
	200	-0.2994	0.0006	0.1029	0.0106	0.0456	2.8867	0.9517
	400	-0.2989	0.0011	0.0746	0.0056	0.0069	3.0097	0.9512
β_{12}	50	0.1996	-0.0004	0.2285	0.0522	0.0268	3.0222	0.9380
	100	0.2012	0.0012	0.1450	0.0210	-0.0239	3.0565	0.9464
	200	0.1986	-0.0014	0.1004	0.0101	0.0142	3.0438	0.9457
	400	0.2017	0.0017	0.0737	0.0054	-0.0612	3.0513	0.9482
σ	50	5.5544	0.8544	1.7439	3.7711	1.2482	4.9153	0.9731
	100	5.0887	0.3887	1.0654	1.2861	1.0453	5.0833	0.9620
	200	4.8842	0.1842	0.6865	0.5053	0.6063	3.6151	0.9570
	400	4.7885	0.0885	0.4691	0.2279	0.4181	3.2190	0.9506

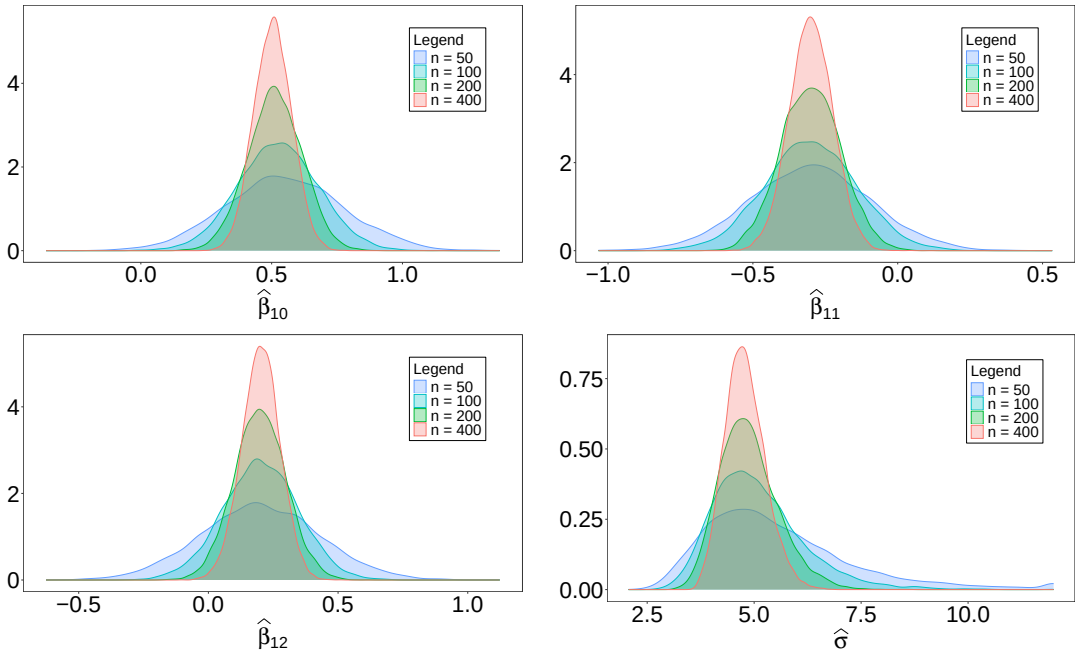
Source: From the authors (2026).

Figure 11: Boxplots of the Monte Carlo simulation results for the estimation of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (0.5, -0.3, 0.2)$ and $\sigma = 4.7$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).
Legend: −: true parameter value; •: mean of the 10,000 replications.

Figure 12: Sampling densities of the Monte Carlo simulation results for the estimation of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (0.5, -0.3, 0.2)$ and $\sigma = 4.7$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).

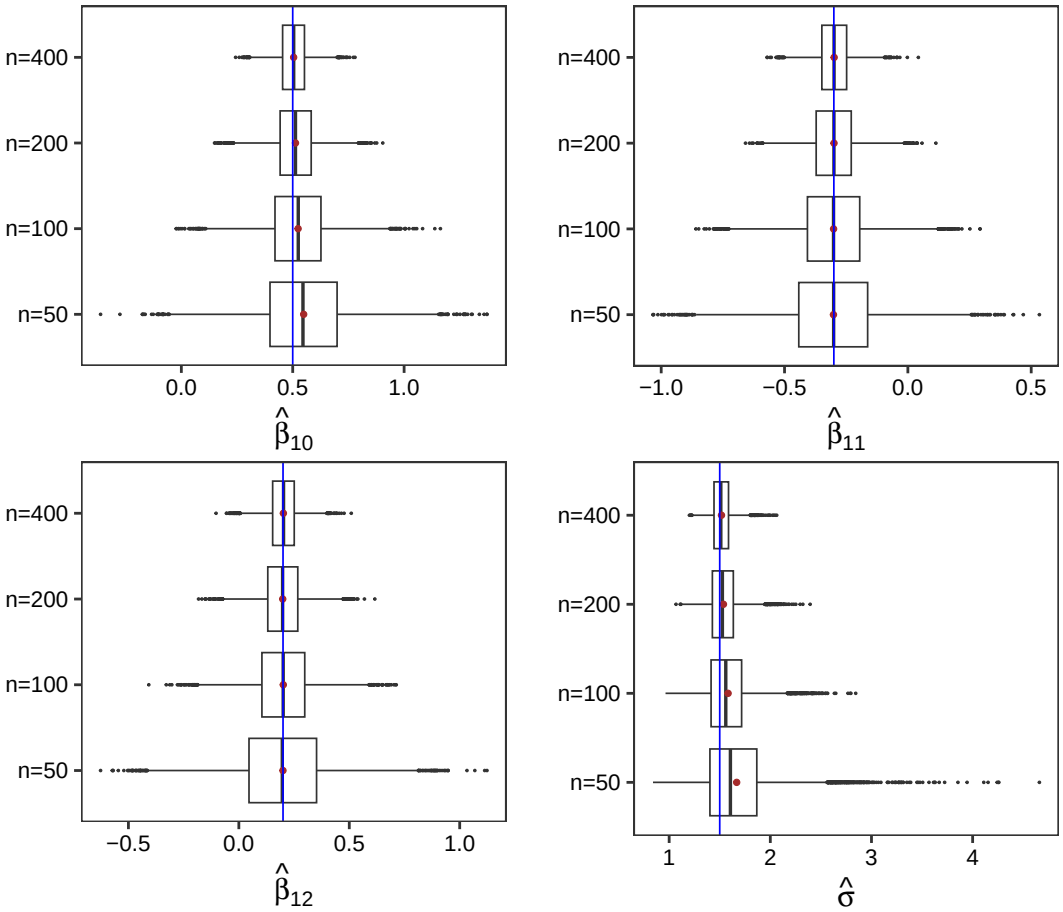
34 ***GSKW model - Scenario:*** $\boldsymbol{\beta}_1 = (1.5, 0.3, 0.2)$ ***and*** $\sigma = 1.5$

Table 7: Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (1.5, 0.3, 0.2)$ and $\sigma = 1.5$, and sample sizes $n \in \{50, 100, 200, 400\}$.

Parameter	n	Mean	Bias	SD	MSE	SC	KC	CP
β_{10}	50	1.5562	0.0562	0.2987	0.0924	0.0396	3.0668	0.9313
	100	1.5288	0.0288	0.2076	0.0439	0.0469	3.0198	0.9431
	200	1.5128	0.0128	0.1469	0.0217	0.0420	3.0259	0.9479
	400	1.5070	0.0070	0.1012	0.0103	0.0231	3.0195	0.9501
β_{11}	50	0.2983	-0.0017	0.3332	0.1110	-0.0079	3.0240	0.9392
	100	0.2982	-0.0018	0.2498	0.0624	-0.0167	3.0988	0.9485
	200	0.3006	0.0006	0.1722	0.0296	0.0013	2.9509	0.9511
	400	0.2998	-0.0002	0.1194	0.0143	0.0075	2.9780	0.9517
β_{12}	50	0.2044	0.0044	0.3720	0.1384	0.0308	3.1644	0.9402
	100	0.2042	0.0042	0.2397	0.0575	-0.0293	2.9840	0.9479
	200	0.2006	0.0006	0.1794	0.0322	-0.0006	3.0410	0.9490
	400	0.1990	-0.0010	0.1197	0.0143	-0.0015	3.0860	0.9512
σ	50	1.6674	0.1674	0.3761	0.1695	1.1698	5.9157	0.9700
	100	1.5827	0.0827	0.2360	0.0625	0.7012	3.8327	0.9579
	200	1.5368	0.0368	0.1574	0.0261	0.5025	3.6034	0.9532
	400	1.5175	0.0175	0.1083	0.0120	0.3538	3.3266	0.9530

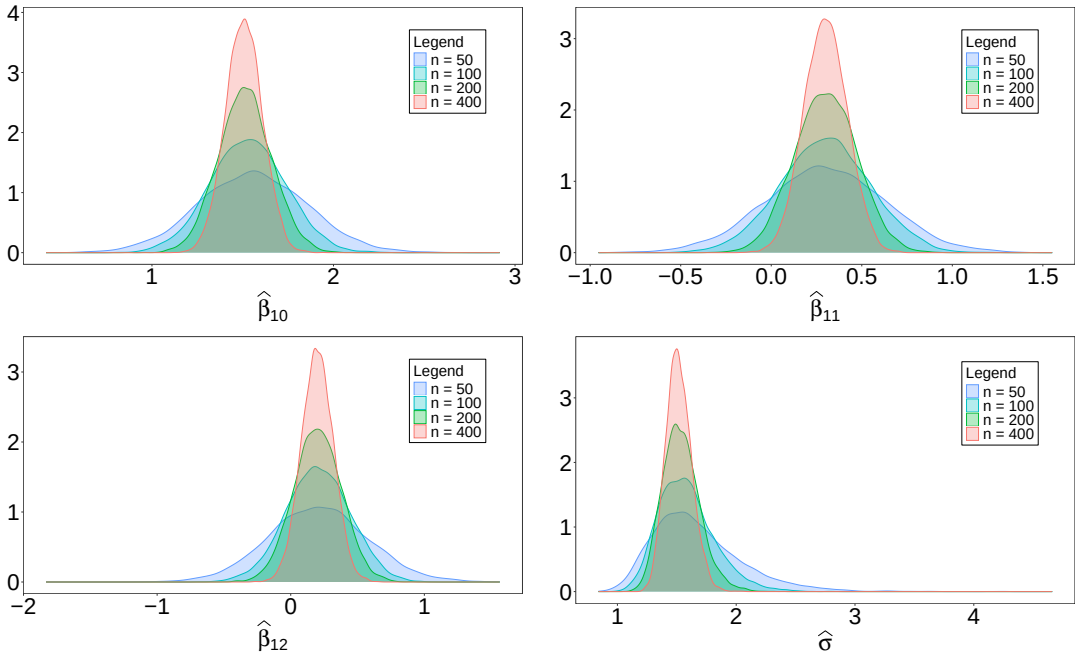
Source: From the authors (2026).

Figure 13: Boxplots of the Monte Carlo simulation results for the estimation of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (1.5, 0.3, 0.2)$ and $\sigma = 1.5$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).
Legend: ▬: true parameter value; •: mean of the 10,000 replications.

Figure 14: Sampling densities of the Monte Carlo simulation results for the estimation of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (1.5, 0.3, 0.2)$ and $\sigma = 1.5$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).

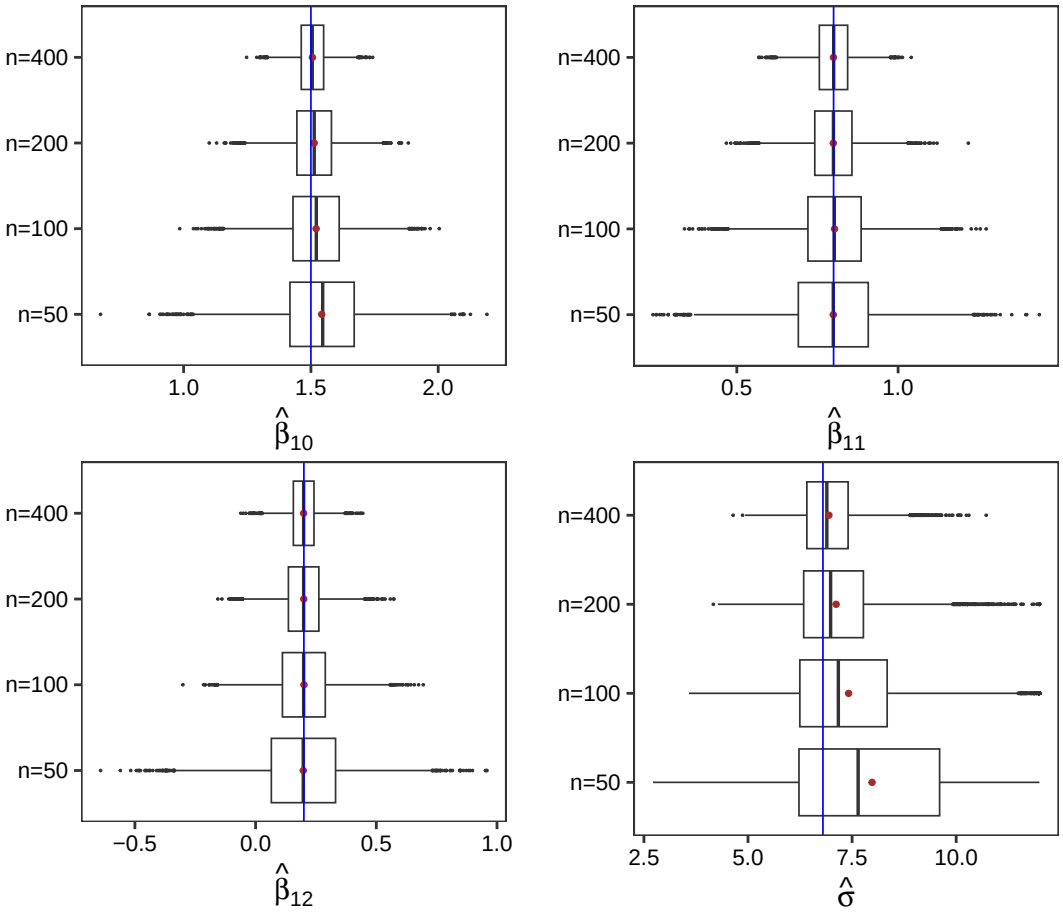
35 ***GSKW model - Scenario:*** $\boldsymbol{\beta}_1 = (1.5, 0.8, 0.2)$ ***and*** $\sigma = 6.8$

Table 8: Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (1.5, 0.8, 0.2)$ and $\sigma = 6.8$, and sample sizes $n \in \{50, 100, 200, 400\}$.

Parameter	n	Mean	Bias	SD	MSE	SC	KC	CP
β_{10}	50	1.5423	0.0423	0.1866	0.0366	-0.1228	2.9417	0.9445
	100	1.5206	0.0206	0.1348	0.0186	-0.0489	2.9595	0.9457
	200	1.5129	0.0129	0.1001	0.0102	-0.0143	2.9865	0.9468
	400	1.5059	0.0059	0.0644	0.0042	0.0043	2.9561	0.9509
β_{11}	50	0.7992	-0.0008	0.1592	0.0253	-0.0080	2.9273	0.9457
	100	0.8030	0.0030	0.1235	0.0153	-0.0094	3.0590	0.9482
	200	0.7992	-0.0008	0.0864	0.0075	0.0018	3.0859	0.9516
	400	0.7996	-0.0004	0.0647	0.0042	-0.0157	2.9286	0.9529
β_{12}	50	0.1976	-0.0024	0.1999	0.0400	-0.0151	3.0662	0.9398
	100	0.2006	0.0006	0.1302	0.0170	0.0420	2.8980	0.9438
	200	0.1995	-0.0005	0.0940	0.0088	0.0438	3.0238	0.9474
	400	0.1989	-0.0011	0.0630	0.0040	-0.0473	3.0320	0.9541
σ	50	7.9805	1.1805	2.2501	6.4568	0.3353	2.1340	0.9702
	100	7.4169	0.6169	1.6335	3.0488	0.7097	3.2520	0.9701
	200	7.1170	0.3170	1.1139	1.3413	0.7067	3.8576	0.9570
	400	6.9439	0.1439	0.7519	0.5860	0.4874	3.4564	0.9513

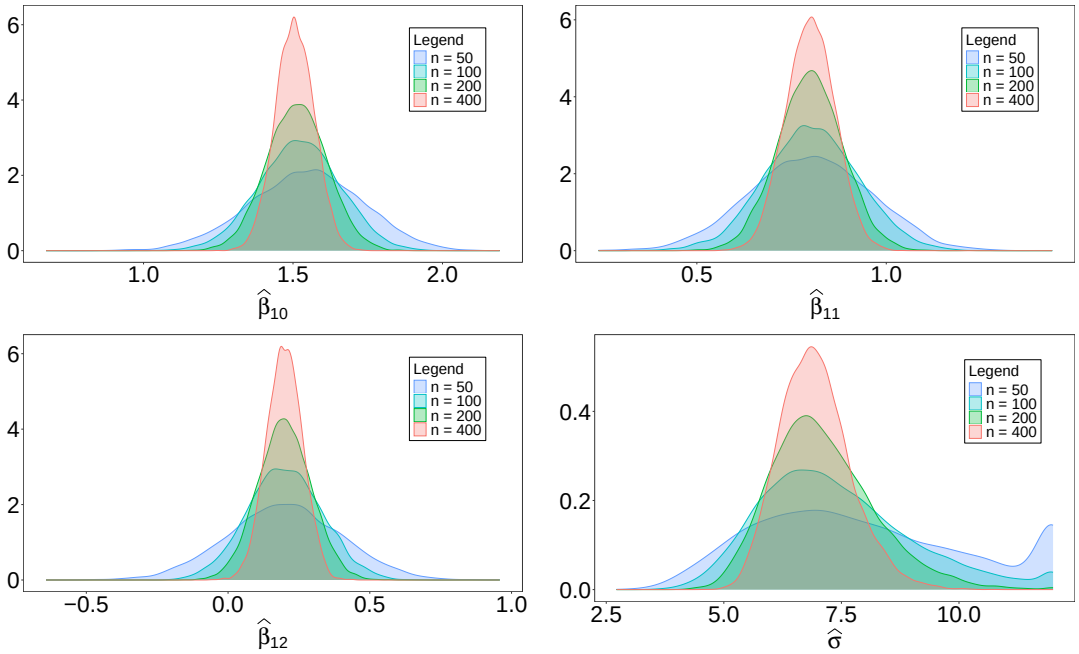
Source: From the authors (2026).

Figure 15: Boxplots of the Monte Carlo simulation results for the estimation of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (1.5, 0.8, 0.2)$ and $\sigma = 6.8$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).
Legend: ▬: true parameter value; •: mean of the 10,000 replications.

Figure 16: Sampling densities of the Monte Carlo simulation results for the estimation of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (1.5, 0.8, 0.2)$ and $\sigma = 6.8$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).

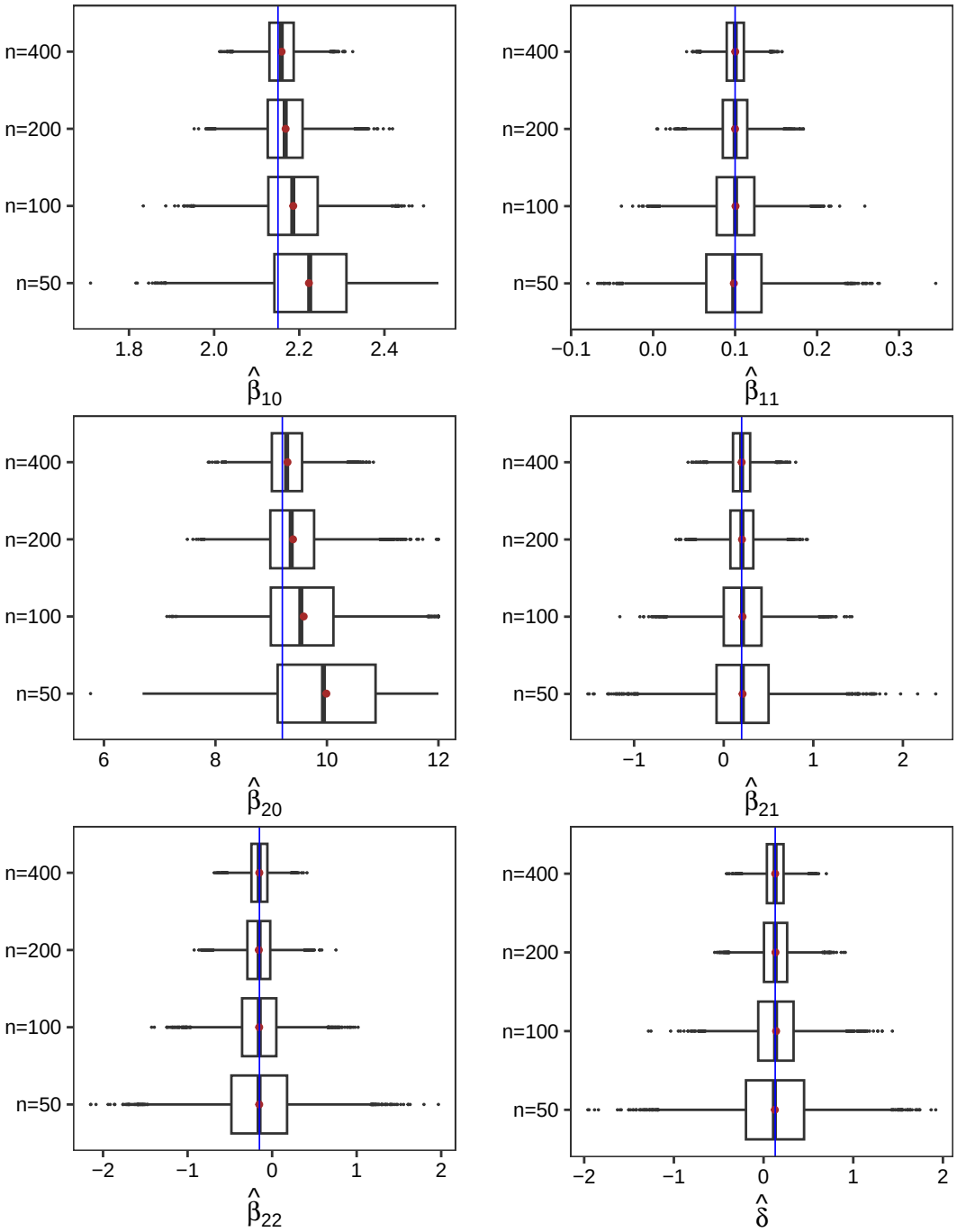
36 ***GSKW model - Scenario:*** $\boldsymbol{\beta}_1 = (2.15, -0.10)$ ***and*** $\boldsymbol{\beta}_2 = (9.20, -0.20, 0.13)$

Table 9: Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (2.15, -0.10)$ and $\boldsymbol{\beta}_2 = (9.20, -0.20, 0.13)$, and sample sizes $n \in \{50, 100, 200, 400\}$.

Parameter	n	Mean	Bias	SD	MSE	SC	KC	CP
β_{10}	50	2.2227	0.0727	0.1185	0.0193	-0.1869	2.5833	0.8997
	100	2.1859	0.0359	0.0863	0.0087	0.0513	2.8789	0.9227
	200	2.1679	0.0179	0.0609	0.0040	0.1101	3.0387	0.9351
	400	2.1585	0.0085	0.0424	0.0019	0.0583	2.9602	0.9430
β_{11}	50	0.0985	-0.0015	0.0492	0.0024	0.0122	3.0391	0.9309
	100	0.1006	0.0006	0.0339	0.0012	0.0006	3.0713	0.9419
	200	0.1000	-0.0000	0.0223	0.0005	0.0112	3.0605	0.9489
	400	0.1002	0.0002	0.0154	0.0002	-0.0127	2.9961	0.9510
β_{20}	50	9.9900	0.7900	1.1874	2.0341	0.0018	2.2555	0.9441
	100	9.5829	0.3829	0.8411	0.8541	0.3104	2.9772	0.9374
	200	9.3920	0.1920	0.5854	0.3796	0.2957	3.1977	0.9414
	400	9.2928	0.0928	0.4055	0.1730	0.1969	3.0163	0.9464
β_{21}	50	0.2106	0.0106	0.4429	0.1962	0.0431	3.2494	0.9393
	100	0.2103	0.0103	0.3138	0.0986	-0.0039	3.0206	0.9472
	200	0.2038	0.0038	0.1902	0.0362	0.0048	3.0093	0.9502
	400	0.2003	0.0003	0.1441	0.0208	0.0172	3.0900	0.9510
β_{22}	50	-0.1524	-0.0024	0.4945	0.2445	0.0080	3.1531	0.9359
	100	-0.1528	-0.0028	0.3063	0.0938	-0.0155	3.1989	0.9450
	200	-0.1572	-0.0072	0.2046	0.0419	-0.0466	3.0498	0.9503
	400	-0.1511	-0.0011	0.1406	0.0198	0.0079	2.9965	0.9512
β_{23}	50	0.1260	-0.0040	0.4863	0.2365	-0.0200	3.1236	0.9387
	100	0.1402	0.0102	0.3006	0.0905	0.0547	3.2263	0.9422
	200	0.1330	0.0030	0.1932	0.0373	0.0197	3.0433	0.9479
	400	0.1300	0.0000	0.1393	0.0194	0.0011	3.0616	0.9487

Source: From the authors (2026).

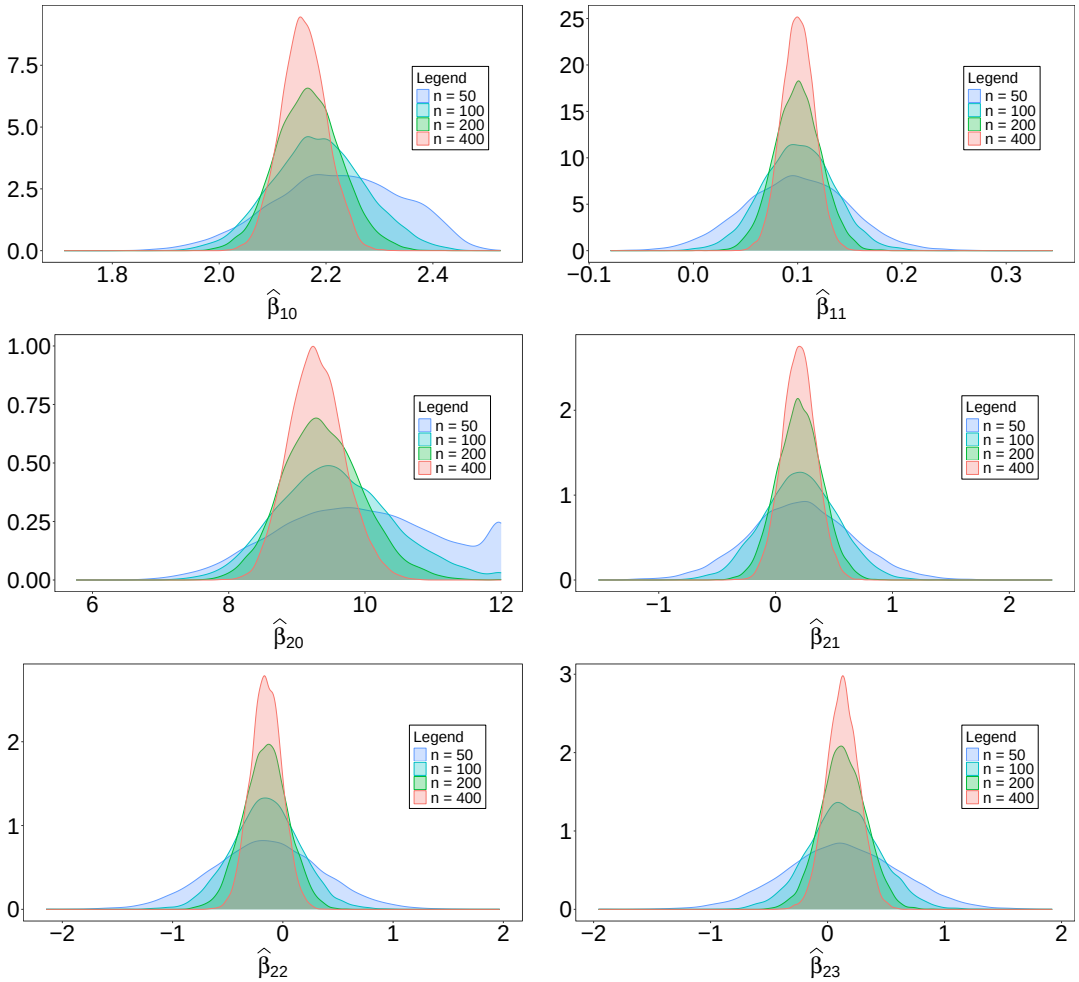
Figure 17: Boxplots of the Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (2.15, -0.10)$ and $\boldsymbol{\beta}_2 = (9.20, -0.20, 0.13)$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).

Legend: ▬: true parameter value; •: mean of the 10,000 replications.

Figure 18: Sampling densities of the Monte Carlo simulation results for the MLE of the parameters of the GSKW model, considering $\boldsymbol{\beta}_1 = (2.15, -0.10)$ and $\boldsymbol{\beta}_2 = (9.20, -0.20, 0.13)$, and sample sizes $n \in \{50, 100, 200, 400\}$.



Source: From the authors (2026).